STACKS, DERIVED CATEGORIES OF SPACES: RESOLUTION-PROPERTY-AMPLE-RELATIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7129-7140 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 7142-7174, the complete author proof (31 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 50 of queue_spaceperfect.json (pb=1368, stmt 7129-7140, proof 7142-7174).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-resolution-property-ample-relative' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01Q3: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7174–7174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D31: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7174–7174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0829: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7174–7174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *Y is quasi-compact and quasi-separated and has the resolution property,*

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *there exists an f-ample invertible module on X (Divisors on Spaces, Definition 0D31).*

Then X has the resolution property.

Proof. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $\mathcal{L}$ be an f-ample invertible module. Choose an affine scheme V and a surjective étale morphism $V \to Y$. Set $U = V \times_Y X$. Then $\mathcal{L}|_U$ is ample on U. By Properties, Proposition 01Q3 we know there exists finitely many maps $s_i : \mathcal{L}^{\otimes n_i}|_U \to \mathcal{F}|_U$ which are jointly surjective. Consider the quasi-coherent $\mathcal{O}_Y$-modules

$$\mathcal{H}_n = f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n})$$

We may think of s_i as a section over V of the sheaf $\mathcal{H}_{-n_i}$. Suppose we can find finite locally free $\mathcal{O}_Y$-modules $\mathcal{E}_i$ and maps $\mathcal{E}_i \to \mathcal{H}_{-n_i}$ such that s_i is in the image. Then the corresponding maps

$$f^*\mathcal{E}_i \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n_i} \longrightarrow \mathcal{F}$$

are going to be jointly surjective and the lemma is proved. By Limits of Spaces, Lemma 0829 for each i we can find a finite type quasi-coherent submodule $\mathcal{H}'_i \subset \mathcal{H}_{-n_i}$ which contains the section s_i over V. Thus the resolution property of Y produces surjections $\mathcal{E}_i \to \mathcal{H}'_i$ and we conclude. □